**Name of the Student: _____Roll no: _____**

**CLASS: - T.E (CSE) Division: Course: - PCC305CSE- Machine Learning Lab**

# Experiment No. 04

**** Use of polynomial regression****

Marks: /10

**Date of Performance:** / /2026 **Sign with Date:**

**Aim:** To Demonstrate the use of polynomial regression.

**Lab Objective**: To observe the effect of model complexity on prediction accuracy in terms of the bias–variance trade-off.

## Problem Statement:

Demonstrate the use of polynomial regression with varying polynomial degrees on a suitable open-access dataset and observe the effect of model complexity on prediction accuracy in terms of the bias–variance trade-off.

## Theory:

### Polynomial Regression Formulation

Polynomial Regression extends linear regression by adding polynomial terms of the feature variables while remaining linear with respect to the parameters (β):

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + ..... + \beta_d X^d + \epsilon$$

Where:

- d is the degree of the polynomial.
- $X, X^2, .... X^d$ are generated using feature transformation techniques (e.g., PolynomialFeatures).

## The Bias–Variance Trade-Off

Total generalization error can be decomposed into:

Total Error = $\text{Bias}^2$ + Variance + Irreducible Error ($\sigma^2$)

- **Underfitting (High Bias, Low Variance):** Occurs at low polynomial degrees (d=1). The model is overly simplistic to capture the underlying nonlinear relationship, resulting in high error on both training and test sets.
- **Overfitting (Low Bias, High Variance):** Occurs at high polynomial degrees (d >= 5). The model fits random noise in the training data, leading to near-zero training error but surging test error.
- **Optimal Balance:** The degree d* where the test Mean Squared Error ($\text{MSE}_{\text{test}}$) reaches its global minimum.

## Recommended Open-Access Dataset

The **Auto MPG Dataset** is an open-access regression benchmark sourced from the **UCI Machine Learning Repository**. It was originally derived from the StatLib library and used in the 1983 American Statistical Association Exposition. The dataset tracks city-cycle fuel efficiency and technical performance metrics for automobile models produced between 1970 and 1982.

- **Total Instances (Rows):** 398 instances (392 after dropping records with missing values)
- **Total Attributes (Columns):** 9 (8 predictors + 1 continuous target)
- **Missing Values:** 6 records in the `horsepower` column contain missing/unknown values (often represented as `?` or `NaN`).
- **Dataset: Auto MPG Dataset** (UCI Machine Learning Repository / `seaborn` built-in)
- **Input Feature (X):** `horsepower` (Engine power)
- **Target Variable (Y):** `mpg` (Miles per gallon - fuel efficiency)
- **Characteristics:** A clear nonlinear (inverse curvature) relationship between engine horsepower and fuel economy.

## Algorithm & Execution Logic

INPUT : Dataset D = {(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)}, Degrees List = [1, 2, 3, 5, 8, 10]
OUTPUT : Train MSE vs. Test MSE curve, Fitted Polynomial Curves, Optimal Degree d*
STEP 1: Load dataset, drop missing/null records, extract feature X and target y.

STEP 2: Split data into 80% Training and 20% Testing subsets.

STEP 3: For each degree d in Degrees List:

3.1 Transform X_train and X_test into polynomial feature spaces of degree d: $X_poly_train = [1, X_train, X_train^2, ..., X_train^d]$ $X_poly_test = [1, X_test, X_test^2, ..., X_test^d]$

3.2 Standardize transformed features using StandardScaler.

3.3 Fit Linear Regression model on scaled polynomial features.

3.4 Predict on both training and testing sets: $\hat{y}_train$, $\hat{y}_test$.

3.5 Calculate and store: $MSE_train = (1 / N_train) * \Sigma (y_train - \hat{y}_train)^2$ $MSE_test = (1 / N_test) * \Sigma (y_test - \hat{y}_test)^2$

STEP 4: Plot Model Complexity Curve (Polynomial Degree vs. MSE_train & MSE_test). STEP 5: Identify d* where MSE_test is minimized (sweet spot of Bias–Variance).

## Questions:

1. What is the fundamental difference between Linear Regression and Polynomial Regression?
2. How does the Bias–Variance trade-off manifest as the polynomial degree increases?
3. Why is feature scaling particularly critical when training high-degree polynomial models?
4. How can you mathematically detect overfitting from the training vs. testing MSE curves.
5. What techniques can be used to prevent overfitting in higher-degree polynomial regression?

| Attendance and Timely Submission (10) | Execution (10) | Observations and Records (10) | Journal Documentation (10) | Oral (10) |
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**Signature of Subject Teacher**